

Web Development Viva Notes

Web Fundamentals · HTML · CSS

Part 1: Quick Revision

Web Fundamentals

- **Internet vs Web:** The Internet is the global network of networks (infrastructure). The Web is a service on it: pages linked via HTTP.
- **Client-server:** The client (browser) sends requests, and the server processes them and responds.
- **Browser:** Sends requests, parses HTML/CSS/JS, builds the DOM, renders the page.
- **Request-response cycle:** URL → DNS lookup → TCP/TLS connection → HTTP request → server response → browser renders.
- **HTTP vs HTTPS:** HTTPS is HTTP encrypted with TLS/SSL (secure, authenticated).
- **Status codes:** 200 OK, 201 Created, 301 Moved, 304 Not Modified, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 500 Server Error.
- **Domain / hosting / server:** The domain is the human-readable address. Hosting is rented space for your files. The server is the machine that serves them.
- **Git vs GitHub:** Git is local version control. GitHub is remote hosting for Git repos. Know init, add, commit, push, pull, clone, branch, merge.

HTML

- **Structure:** `<!DOCTYPE html>`, `<html>`, `<head>` (metadata), `<body>` (content).
- **Lists:** `ul`, `ol`, `dl`. **Tables:** `table`, `tr`, `th`, `td`, plus `thead/tbody`, `colspan/rowspan`.
- **Forms:** form with action and method. Input types: text, email, password, number, radio, checkbox, date, file, submit. Always pair label for with id.
- **Semantic elements:** header, nav, main, section, article, aside, footer. They improve SEO, accessibility and readability.
- **Accessibility:** alt text, labels, semantic tags, heading order, keyboard navigation, contrast, ARIA only when needed.

CSS

- **Selectors:** element, class (`.`), id (`#`), descendant, child (`>`), group, pseudo-class (`:hover`). Specificity: inline > id > class > element.
- **Box model:** content + padding + border + margin. box-sizing: border-box includes padding and border in the width.
- **Positioning:** static, relative, absolute (relative to nearest positioned ancestor), fixed (viewport), sticky.
- **Flexbox:** 1D layout. Container: `display:flex`, `flex-direction`, `justify-content` (main axis), `align-items` (cross axis), `flex-wrap`, `gap`. Items: `flex: grow shrink basis`.
- **Grid:** 2D layout. `display:grid`, `grid-template-columns: repeat(3, 1fr)`, `gap`, `grid-area`.
- **Responsive design:** Layouts adapt to screen size using fluid units (`%`, `rem`, `vw`), flexible images and media queries.
- **Mobile-first:** Write base styles for mobile, then enhance with `@media` (min-width: 768px).
- **Media queries:** `@media (max-width: 600px) { ... }`. Also add the viewport meta tag: `width=device-width, initial-scale=1`.

Part 2: Core Viva Questions

1. Flexbox vs Grid?

Flexbox is one-dimensional: it lays items out in a single row or column. Grid is two-dimensional: it controls rows and columns together. Use Flexbox for navbars and aligning items in a line; use Grid for full page layouts and card galleries.

2. absolute vs relative?

- **relative:** positioned relative to its own normal position and still occupies its original space in the flow.
- **absolute:** removed from normal flow and positioned relative to its nearest positioned ancestor (not static). If none exists, it uses the page.
- Common pattern: parent position: relative, child position: absolute.

3. Why use semantic tags?

- Give meaning to the structure (nav, article, footer vs generic div).
- Better accessibility: screen readers can navigate by landmarks.
- Better SEO: search engines understand the content.
- Cleaner, more readable, easier-to-maintain code.

4. GET vs POST?

- **GET:** requests data. Data goes in the URL (visible, length-limited), can be bookmarked and cached. Not for sensitive data.
- **POST:** sends data to the server. Data goes in the request body (not shown in the URL), no practical size limit, not cached. Used for login, signup, uploads.

5. What happens when you type a URL?

- 1 The browser checks its cache for the address.
- 2 DNS lookup converts the domain to an IP address.
- 3 A TCP connection is established (plus a TLS handshake for HTTPS).
- 4 The browser sends an HTTP request (e.g., GET /).
- 5 The server processes it and returns a response (status code, headers, HTML).
- 6 The browser parses the HTML, builds the DOM, fetches CSS, JS and images, and renders the page.

Part 3: Mock Viva Questions

1. What is the box model?

Every element is a box made of content, padding, border and margin (inside to outside). box-sizing: border-box makes width include padding and border.

2. Difference between id and class?

An id is unique per page (#id), while a class can be reused on many elements (.class). Id has higher specificity.

3. margin vs padding?

Padding is space inside the border, between content and border. Margin is space outside the border, between elements.

4. What are relative, fixed and sticky?

Fixed stays in place relative to the viewport while scrolling. Sticky behaves as relative until a scroll threshold, then sticks like fixed. Relative offsets from its normal position.

5. Block vs inline elements?

Block elements (div, p, h1) start on a new line and take full width. Inline elements (span, a) flow within text and only take the width they need. Width and height don't apply to inline elements.

6. Difference between div and section?

div is a generic container with no meaning. section is semantic and groups related content, usually with a heading.

7. What is the viewport meta tag and why is it needed?

It tells mobile browsers to match the page width to the device width. Without it, mobile browsers render a zoomed-out desktop layout and media queries don't behave as expected.

8. justify-content vs align-items?

justify-content aligns items along the main axis. align-items aligns them along the cross axis. In a row, main is horizontal and cross is vertical.

9. Difference between Internet and Web?

The Internet is the global network infrastructure. The Web is a service running on it, made of linked pages accessed via HTTP/HTTPS.

10. 404 vs 500 error?

404 is a client-side error: the requested resource was not found. 500 is a server-side error: something went wrong while processing a valid request.

Bonus: Git vs GitHub?

Git is the version control tool. GitHub is a cloud platform for hosting Git repositories and collaborating.

Bonus: git pull vs git clone?

Clone copies a whole remote repo for the first time. Pull fetches and merges new changes into an existing local repo.